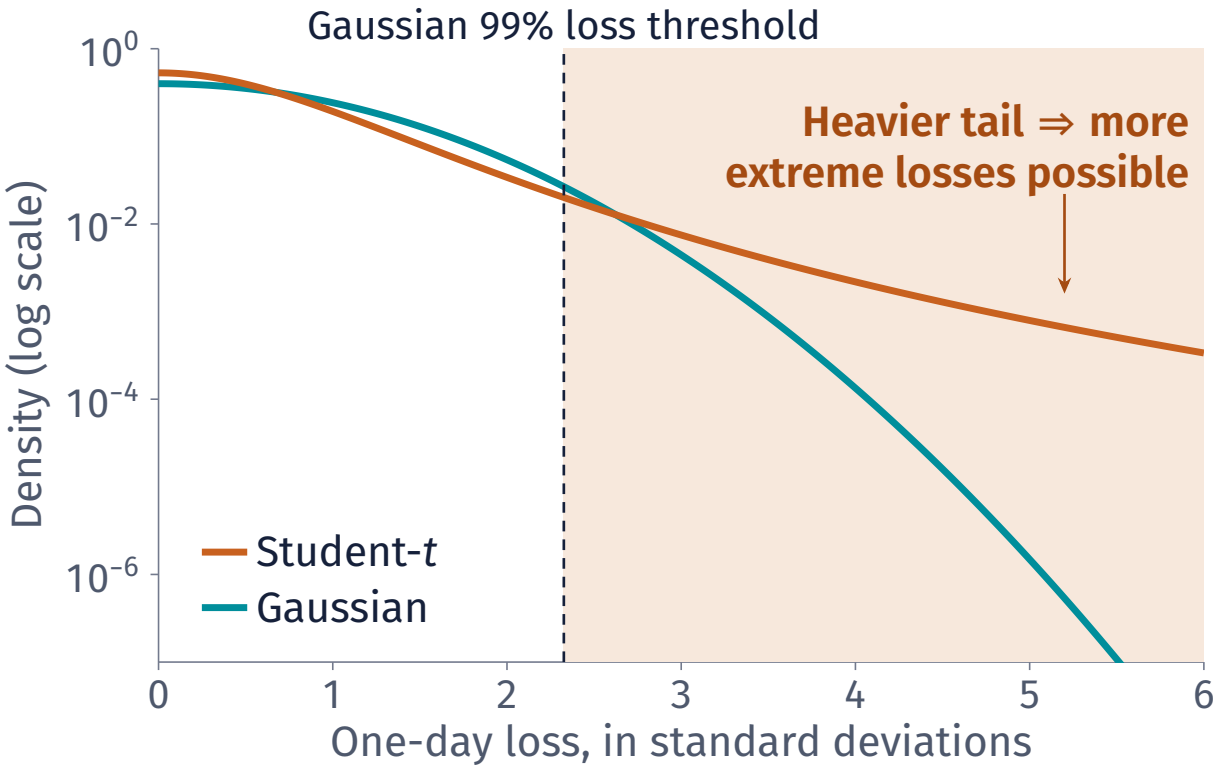
How much risk hides beyond the Gaussian tail?

Bayesian estimation of one-day 99% VaR & Expected Shortfall



Risk limits depend on the worst 1% of days.

How heavy that tail is must be learned from limited data.

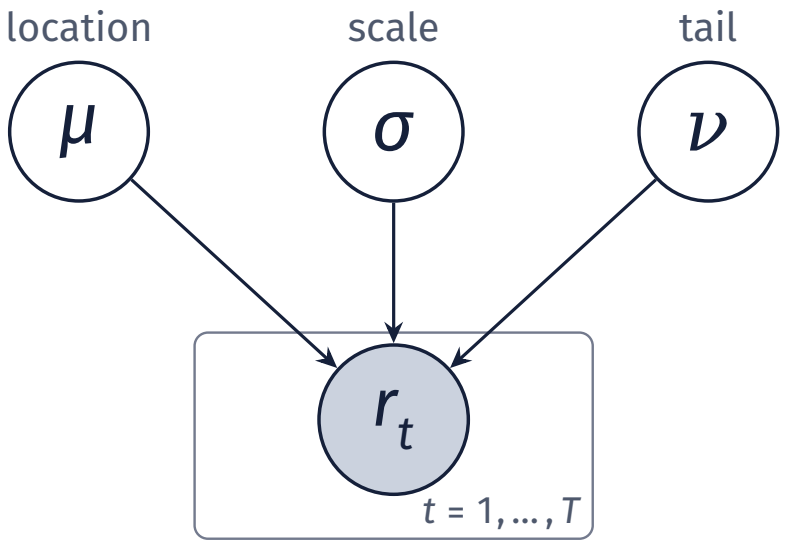
How do heavy tails and parameter uncertainty change our estimates of extreme one-day losses?

Synthetic distribution comparison — not market data. Student-t with $\nu = 4$; both curves have unit variance.

VaR_{99} the loss exceeded on only 1% of days **ES_{99}** the average loss on those days

A small Bayesian model for financial tail risk

Three unknowns, one observed series



$$r_t \mid \mu, \sigma, \nu \overset{\text{iid}}{\sim} \text{Student-}t_\nu(\mu, \sigma)$$

Days are independent given (μ, σ, ν):
no volatility clustering (a stated limitation).

Data

$r_t = \log(P_t / P_{t-1})$, daily

Broad equity index (e.g. S&P 500). First: simulated data with known parameters.

Priors

$\mu \sim \text{N}(0,\ 0.01^2)$

$\sigma \sim \text{HalfNormal}(0.02)$

$\nu - 2 \sim \text{Exp}(0.1)$

Weakly informative in daily units; sensitivity tested.

Target

$p(\mu, \sigma, \nu \mid r_{1:T}) \rightarrow$ future loss $L = -r_{T+1}$

Known only up to a constant ⇒ MCMC.

Method & difficulties

From posterior simulation to a defensible risk estimate

Our own sampler, checked on simulated data before real data

1 **Simulate** → data with known values

2 **Sample** → our own MH sampler

3 **Check** → recovery & convergence

4 **Estimate** → 99% VaR & ES, real data

The sampler

- Random-walk Metropolis–Hastings, coded by us
- on $(\mu,\ \log\sigma,\ \log(\nu - 2))$, Jacobian included
- proposal tuned in pilot runs, then frozen

The checks

- true parameters inside 95% credible intervals
- 4 chains: trace, $\hat{R}$, ESS, MC error
- predictive tail checks, prior sensitivity

Output: 99% VaR & ES, Gaussian plug-in vs Bayesian Student-t, with 95% credible intervals.

Expected difficulty: σ and ν trade off ⇒ slow mixing; ES rests on the worst 1% ⇒ Monte Carlo error.